■ `Graphics3D`

`Graphics3D[`*primitives*`,` *options*`]` represents a three-dimensional graphical image.

`Graphics3D` is displayed using `Show`. ■ The following primitives can be used:

`Point[{`*x*`,` *y*`,` *z*`}]`	point
`Line[{{`x_1`,` y_1`,` z_1`},` ...`}]`	line
`Polygon[{{`x_1`,` y_1`,` z_1`},` ...`}]`	polygon
`GrayLevel[`*i*`]`	gray level specification
`RGBColor[`*r*`,` *g*`,` *b*`]`	color specification
`PointSize[`*s*`]`	point size specification
`Thickness[`*t*`]`	line thickness specification
`FaceForm[`*spec*`]`	polygon face specification
`EdgeForm[`*spec*`]`	polygon edge specification

■ The following options can be given:

`AmbientLight`	`GrayLevel[0.]`	ambient illumination level
`AspectRatio`	`1`	ratio of height to width
`BoxRatios`	`Automatic`	bounding 3D box ratios
`Boxed`	`True`	whether to draw the bounding box
`DisplayFunction`	`$DisplayFunction`	function for generating output
`Framed`	`False`	whether to draw a frame
`LightSources`	(see below)	positions and colors of light sources
`Lighting`	`True`	whether to use simulated illumination
`Plot3Matrix`	`Automatic`	perspective transformation matrix
`PlotColor`	`True`	whether to plot in color
`PlotLabel`	`None`	a label for the plot
`PlotRange`	`Automatic`	range of values to include
`RenderAll`	`True`	whether to render all polygons
`Shading`	`True`	whether to shade polygons
`ViewPoint`	`{1.3, -2.4, 2.}`	viewing position

■ Nested lists of graphics primitives can be given. The scope of specifications such as `GrayLevel` is the list that contains them. ■ The standard print form for `Graphics3D[`...`]` is `-Graphics3D-`. `InputForm` prints the explicit list of primitives. ■ The default light sources used are `{{{1,0,1}, RGBColor[1,0,0]}, {{1,1,1}, RGBColor[0,1,0]}, {{0,1,1}, RGBColor[0,0,1]}}` ■ See page 148. ■ See also: `Plot3D`, `SurfaceGraphics`.

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